

BTP PROJECT PROPOSAL

Project Title -

AI-Assisted CFD Optimization and Thermodynamic Analysis of a Heat-Integrated Oscillatory Baffled Reactor with Perforated Twisted Baffles for Process Intensification

1. Introduction

Efficient mixing and heat transfer are important requirements in modern chemical and thermal engineering systems. Conventional tubular reactors and heat exchangers often suffer from **poor mixing, non-uniform temperature distribution, formation of dead zones, and high energy consumption.**

These limitations reduce reactor efficiency and increase operational cost.

To overcome these problems, advanced reactor systems known as **Oscillatory Baffled Reactors (OBRs)** are increasingly being studied because they provide enhanced mixing and heat transfer even at low flow rates.

This project proposes **the design and optimization of a Heat-Integrated Oscillatory Baffled Reactor (OBR) containing perforated twisted baffles and a central heat-transfer tube.** The project combines concepts of:

- oscillatory flow,
- vortex generation,
- heat transfer enhancement,
- CFD simulation,
- optimization techniques,
- and AI-assisted predictive modeling.

The proposed reactor aims to improve thermo-hydraulic performance and achieve process intensification while maintaining acceptable pressure losses.

2. Problem Statement

Conventional reactors and heat exchangers face several engineering challenges such as:

- poor fluid mixing,
- low heat transfer efficiency,
- formation of dead zones,
- high pumping power requirements,
- non-uniform temperature distribution,
- and large equipment size.

Although oscillatory baffled reactors and helical reactors already exist, optimization of internal baffle geometry and thermo-hydraulic performance remains an important research area.

This project focuses on improving reactor performance using:

- **perforated twisted baffles**
 - **oscillatory flow**
 - **AI-assisted CFD optimization techniques.**
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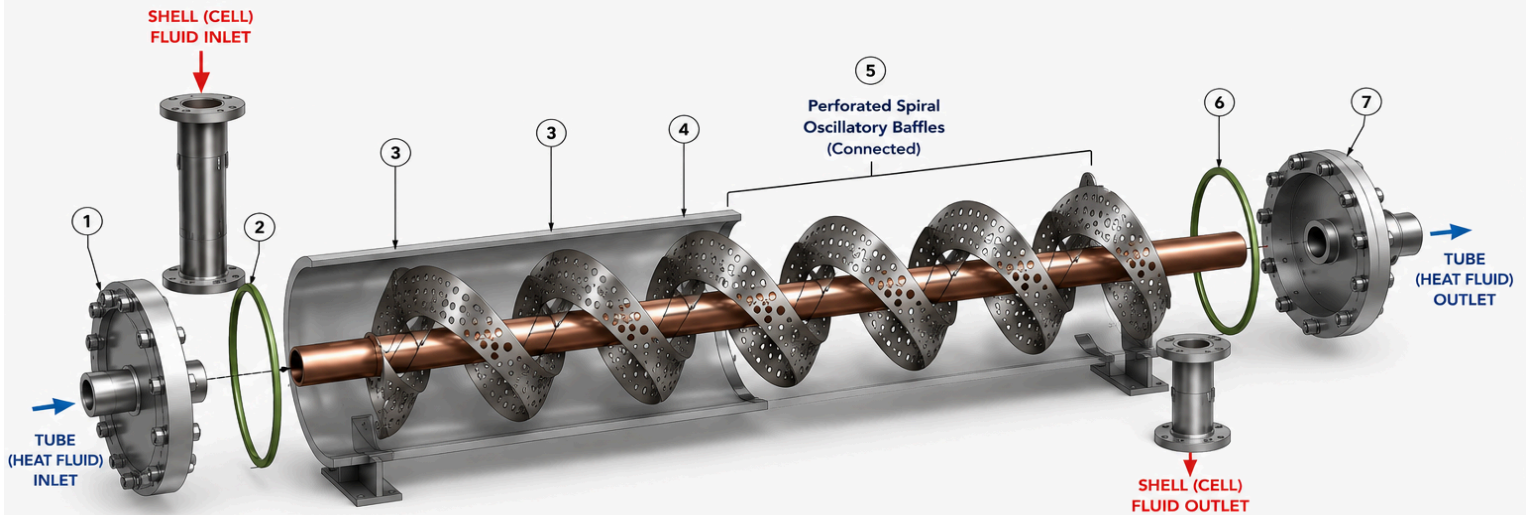
3. Aim of the Project

To design, simulate, optimize, and thermodynamically analyze a Heat-Integrated Oscillatory Baffled Reactor using **CFD**, **RSM optimization**, and **ANN-based prediction models** for enhanced heat transfer and process intensification.

5. Proposed Reactor Design

The proposed reactor consists of:

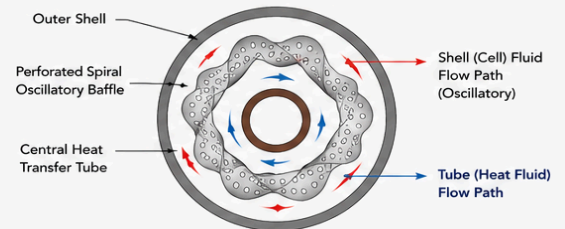
EXPLODED VIEW OF HEAT-INTEGRATED HYBRID OSCILLATORY BAFFLED REACTOR (PERFORATED SPIRAL BAFFLES)



No.	Component	Description
1	Inlet End Cover	Provides sealed entry for tube (heat fluid)
2	Inlet Gasket / Seal	Ensures leak-proof connection
3	Outer Shell (Cell)	Contains shell-side (cell) fluid
4	Central Heat Transfer Tube	Carries heating / cooling fluid
5	Perforated Spiral Oscillatory Baffles (Connected)	Generate oscillatory flow, enhance mixing, and improve heat transfer
6	Outlet Gasket / Seal	Ensures leak-proof connection
7	Outlet End Cover	Provides sealed exit for tube (heat fluid)

Note: Spiral baffles are fixed (not mixed) and connected along the entire length.

CROSS-SECTIONAL VIEW (FRONT) (AT ANY SECTION)



5.1 Perforated Twisted Baffles

Perforated twisted baffles are mounted periodically around the central tube.

Purpose:

- vortex generation,
- enhanced mixing,
- turbulence enhancement,
- improved heat transfer,
- reduction of stagnant regions.

5.2 Oscillatory Flow Mechanism

Oscillatory flow is introduced using sinusoidal inlet velocity conditions.

Purpose:

- periodic vortex generation,
 - enhanced mixing at low Reynolds number,
 - improved residence time distribution.
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6. Methodology

Stage 1 — Literature Review and Problem Identification

A detailed literature review will be conducted on:

- oscillatory baffled reactors,
- heat-integrated reactors,
- shell-and-tube thermal systems,
- perforated baffle configurations,
- CFD-based thermal-fluid analysis,
- process intensification,
- entropy generation analysis,
- and AI-assisted optimization methods.

Research papers from journals related to:

- heat transfer,
- CFD,
- chemical engineering,
- and reactor optimization

will be studied to identify:

- existing limitations,
- important parameters,
- governing physics,
- and optimization strategies.

This stage will help finalize:

- reactor dimensions,
 - operating conditions,
 - and simulation parameters.
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Stage 2 — Reactor Design, CFD Simulation and Thermo-Hydraulic Analysis

1. The reactor will be designed using SolidWorks.
 2. The geometry will be meshed using **ANSYS Meshing**, followed by mesh refinement and mesh-independence studies.
 3. Important geometrical and operating parameters such as **Reynolds number, oscillation frequency, twist angle, perforation diameter, baffle spacing, and porosity** will be varied.
 4. CFD simulations will be performed in **ANSYS Fluent** under oscillatory flow conditions.
 5. The following performance parameters will be evaluated:
 - Nusselt number,
 - heat transfer coefficient,
 - pressure drop,
 - thermo-hydraulic performance factor,
 - and mixing index.
 6. The **CFD results** will be collected as a structured dataset, with each simulation case containing the corresponding input parameters and performance outputs for subsequent **DOE, RSM, and ANN analysis**.
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Stage 3 — Response Surface Methodology (RSM)

1. The CFD-generated dataset will then be analyzed using **Response Surface Methodology (RSM)** in Python.
 2. RSM will be used to determine the effects and interactions of reactor parameters and identify optimum operating conditions.
 3. RSM will be used to:
 - establish mathematical relationships,
 - generate regression equations,
 - create response surfaces,
 - and identify optimum operating conditions.
 4. Contour plots and 3D response surface plots will be generated to study the interaction between different parameters.
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Stage 4 — ANN-Based Prediction Modeling

1. The CFD-generated dataset will be used to develop an **Artificial Neural Network (ANN)** model in Python.
2. The ANN model will learn the relationship between:
 - reactor geometry,
 - operating conditions,
 - and reactor performance.
3. **ANN Inputs** will include:
 - Reynolds number (**Re**),
 - oscillation frequency (**f**),
 - twist angle (**θ**),
 - perforation diameter (**d_p**),
 - baffle spacing (**S**),

- porosity (ϵ)
4. **ANN Outputs** will include:
 - Nusselt number (**Nu**),
 - pressure drop (**ΔP**),
 - and mixing index (**MI**).
 5. The dataset will be divided into:
 - training data,
 - validation data,
 - and testing data.
 6. The ANN model will be trained and validated using the CFD simulation results.
 7. ANN performance will be evaluated using:
 - **R² (coefficient of determination)**,
 - **RMSE (Root Mean Square Error)**,
 - **MAE (Mean Absolute Error)**,
 - and predicted-versus-CFD comparison plots.
 8. The trained ANN model will provide **rapid prediction of reactor performance** for new operating conditions, reducing the need for additional computationally expensive CFD simulations.
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Stage 5 — Entropy Generation Analysis

Entropy generation analysis will be performed for thermodynamic evaluation of the reactor system.

The analysis will include:

- thermal entropy generation,
- frictional entropy generation,
- and Bejan number calculation.

This stage will help identify:

- energy losses,
 - thermal irreversibility,
 - and thermodynamically efficient operating conditions.
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Stage 6 — Final Performance Evaluation

1. The **optimized reactor configuration** obtained from the RSM analysis will be selected for final evaluation.
2. A **final CFD simulation** of the optimized configuration will be performed to verify the predicted performance.
3. The final CFD results will be compared with the:
 - RSM predictions,
 - ANN predictions,
 - and baseline/conventional reactor results.
4. The reactor performance will be evaluated based on:
 - **Nusselt number (Nu),**
 - **heat-transfer coefficient (h),**
 - **pressure drop (ΔP),**
 - **mixing index (MI),**
 - **thermo-hydraulic performance factor,**
 - and **entropy generation.**
5. Flow and thermal characteristics such as:
 - velocity distribution,
 - temperature distribution,
 - vortex formation,
 - pressure distribution,
 - and mixing behavior

will be compared between the baseline and optimized configurations.

6. The **accuracy of the ANN and RSM models** will be assessed by comparing their predictions with the final CFD results.
 7. The final reactor configuration will be selected based on:
 - enhanced heat-transfer performance,
 - improved mixing,
 - acceptable pressure drop,
 - improved thermo-hydraulic efficiency,
 - and reduced thermodynamic losses.
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8. Software and Tools

Purpose	Software
Cad Modeling	Solidworks
CFD Simulation	Ansys Fluent
Meshing	Ansys Meshing
RSM	Python
ANN Modeling	Python
Data-processing & Visualization	Python

9. References

1. [Design and simulations of a helical oscillatory baffled reactor for biochemical reactions](#)
2. [Development of a more robust correlation for predicting heat transfer performance in oscillatory baffled reactors](#)
3. [Design, performance characterization and applications of continuous oscillatory baffled reactors](#)
4. [Numerical study of structural parameters of perforated baffle on heat transfer enhancement in coiled elastic copper tube heat exchanger](#)